RIP OSPF DHCP NAT ACL VPN IPSec

GLOBAL MULTI-NATIONAL COMPANY · INFORMATION TECHNOLOGY DIVISION

Network Infrastructure

Enterprise-Level Network Design Report

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Contents

1 Executive Summary **2**

2 Network Topology & Architecture **3**
2.1 Overview of the Three-Zone Architecture . . . 3
2.2 Network Topology Diagrams . . . 3
2.2.1 Zone Breakdown . . . 5
2.2.2 IP Addressing Scheme . . . 6

3 Configuration, Results & Testing **7**
3.1 RIP v2 — Routing Information Protocol . . . 7
3.1.1 What is RIP? . . . 7
3.1.2 RIP Configuration (Cisco IOS) . . . 7
3.1.3 Verification . . . 8
3.2 DHCP — Dynamic Host Configuration Protocol . . . 8
3.2.1 What is DHCP? . . . 8
3.2.2 DHCP Pool Configuration . . . 8
3.2.3 Verification . . . 9
3.3 NAT — Network Address Translation . . . 9
3.3.1 What is NAT? . . . 9
3.3.2 NAT Configuration — Router 5 . . . 9
3.3.3 Translation Table (Verified) . . . 10
3.4 ACL — Access Control Lists . . . 10
3.4.1 What are ACLs? . . . 10
3.4.2 ACL Implementation — Router 7 . . . 10
3.4.3 Verification . . . 11
3.5 VPN — Virtual Private Network . . . 11
3.5.1 What is a VPN Tunnel? . . . 11
3.5.2 Verification — Traceroute . . . 11

4 Security Architecture Analysis **13**
4.1 Defense-in-Depth Model . . . 13
4.2 Threat Mitigation Summary . . . 14

5 Results & Conclusion **15**
5.1 End-to-End Testing Summary . . . 15
5.2 Conclusion . . . 15

1 Executive Summary

This report documents the complete design, configuration, and validation of an enterprise-grade network for a **Global Multi-National Company (GMNC)** spanning three geographically distributed zones, supporting hundreds of end-devices, multiple server clusters, and secure inter-site communication.

Project Scope at a Glance

Organization	Global Multi-National Company (GMNC)
Designed By	Saidinesh Andekar — Networking & Cybersecurity Enthusiast
Date	April 6, 2026
Simulator	Cisco Packet Tracer
Zones	Access Zone · Core Operations Zone · Distribution Zone
Technologies	RIP v2, OSPF, DHCP, NAT, ACL, IPSec VPN
IP Scheme	192.168.0.0/16 (subnetting per department)

The network implements protocols analogous to those used by **Amazon AWS**, **Google**, and **Tata Consultancy Services**, ensuring scalability, redundancy, and security at every layer.

2 Network Topology & Architecture

2.1 Overview of the Three-Zone Architecture

The network is divided into three functional zones, mirroring real-world enterprise campus designs used by companies like **Infosys** or **Microsoft**.

Real-World Analogy: Corporate Campus

Think of it like a university campus:

- **Access Zone** = Student labs and dorms — end-users, PCs, laptops connect here
- **Core Operations Zone** = The server room / data center — critical infrastructure
- **Distribution Zone** = Backbone corridors — high-speed inter-router links

2.2 Network Topology Diagrams

The diagrams below show the full logical topology as built in Cisco Packet Tracer.

TOPOLOGY

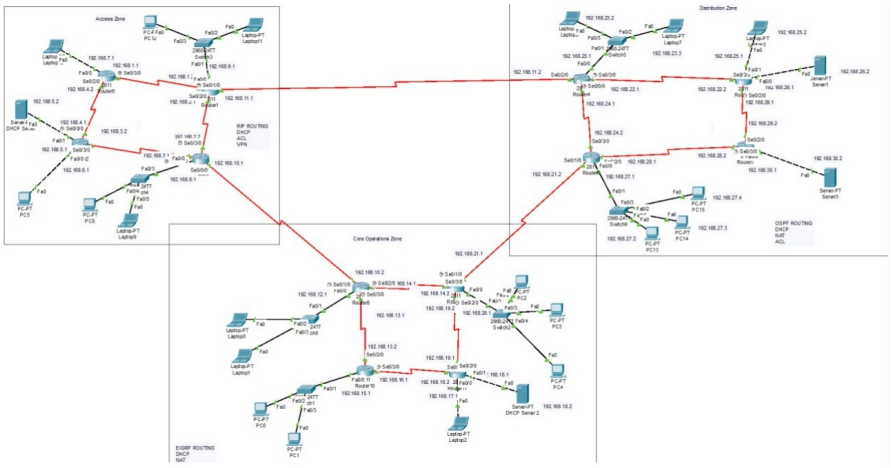


Figure 1: Logical Network Schematic — Cisco Packet Tracer *Access · Core · Distribution Zones*

Key Insight

Zone Colour Key (Figure 2): **Green** = Access Zone **Blue** = Core Operations **Pink** = Distribution Zone

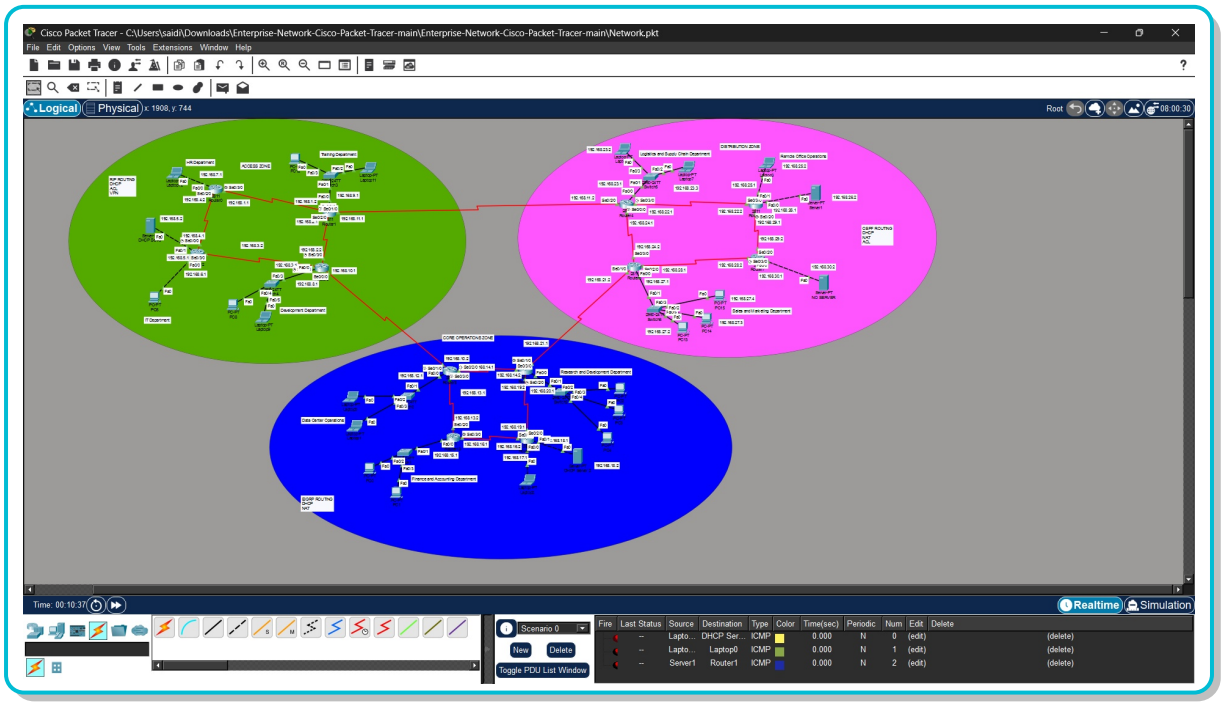


Figure 2: Colour-Coded Zone Overview — Packet Tracer Realtime Simulation *Scenario 0 · Realtime Mode*

2.2.1 Zone Breakdown

Zone	Devices	Function
Access Zone	PCs, Laptops, DHCP Server, L2 Switches	End-user connectivity
Core Operations	Core Routers, OSPF backbone, inter-zone links	High-speed packet forwarding
Distribution	Distribution Routers, ACL, NAT gateway	Policy control & internet access

2.2.2 IP Addressing Scheme

Key Insight

Each department gets its own `/24` subnet, preventing broadcast storms and simplifying troubleshooting — the same approach used by **Amazon's internal DC networks**.

Subnet	Zone / Area	Purpose
`192.168.1.0/24`	Access Zone – Area 1	End-user devices
`192.168.3.0/24`	Access Zone – Area 1	Server VLAN
`192.168.5.0/24`	Access Zone – Area 1	Wireless segment
`192.168.9.0/24`	Access Zone – Area 3	DHCP auto-assigned
`192.168.21.0/24`	Core Operations Zone	Backbone links
`192.168.25.0/24`	Distribution Zone	Inter-router WAN
`192.168.26.0/24`	Distribution Zone	Internet-facing

3 Configuration, Results & Testing

3.1 RIP v2 — Routing Information Protocol

3.1.1 What is RIP?

RIP v2 is a **distance-vector routing protocol** where routers share their entire routing table with neighbours every 30 seconds, using **hop count** as metric (max 15 hops).

Real-World Analogy: Word-of-Mouth Navigation

Lost in a new city, you ask for directions. Each person only knows the next step — "Go to Street A, then ask again." Every router knows one hop ahead and relays info forward until everyone knows every route.

Real usage: Small ISPs and regional bank branch offices still use RIP for its simplicity.

3.1.2 RIP Configuration (Cisco IOS)

```
Router> enable
Router# configure terminal
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.1.0
Router(config-router)# network 192.168.3.0
Router(config-router)# network 192.168.5.0
Router(config-router)# no auto-summary
Router(config-router)# end
Router# show ip route rip
```

Listing 3.1: RIP v2 on Access Zone Router

3.1.3 Verification

`show ip route rip` confirmed **30 networks** (192.168.1.0–192.168.30.0) learned across all routers via Serial10/3/0 and Serial10/1/0.

Key Insight

Limitation: RIP's 15-hop ceiling is unsuitable for very large networks. Enterprises like **Deutsche Bank** use OSPF/BGP internally, reserving RIP only for legacy branches.

3.2 DHCP — Dynamic Host Configuration Protocol

3.2.1 What is DHCP?

DHCP automatically assigns IP addresses, subnet masks, gateways, and DNS to devices on join — no manual configuration needed.

Real-World Analogy: Hotel Reception

Check-in → reception hands you a room number (IP), key card (gateway), WiFi password (DNS). You don't choose — assigned automatically, returned on checkout.

Real usage: Every home router (Jio, Airtel, BSNL) runs DHCP. **Wipro's** 200,000-seat network uses centralised DHCP servers per zone.

3.2.2 DHCP Pool Configuration

```
Router(config)# ip dhcp pool router3pool
Router(dhcp-config)# network 192.168.8.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.8.1
Router(dhcp-config)# dns-server 8.8.8.8

Router(config)# ip dhcp pool router1pool
```

```
Router(dhcp-config)# network 192.168.9.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.9.1

Router(config)# ip dhcp pool serverPool
Router(dhcp-config)# network 192.168.6.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.6.1
```

Listing 3.2: DHCP Server Pools

3.2.3 Verification

PC12 auto-received: IP `192.168.9.2`, Mask `255.255.255.0`, Gateway `192.168.9.1`.

3.3 NAT — Network Address Translation

3.3.1 What is NAT?

NAT translates private internal IPs (192.168.x.x) to a public IP for internet traffic, conserving IPv4 space and hiding internal topology.

Real-World Analogy: Corporate PO Box

500 employees, one public postal address. Outgoing mail stamped with company address; replies arrive at reception and are routed internally. NAT does this for IP packets.

Real usage: Jio/Airtel use **Carrier-Grade NAT** for millions of users. AWS EC2 private subnets use NAT Gateways for internet access.

3.3.2 NAT Configuration — Router 5

```
Router5(config)# interface FastEthernet0/1
Router5(config-if)# ip nat inside
Router5(config-if)# exit
Router5(config)# interface FastEthernet0/0
Router5(config-if)# ip nat outside
Router5(config-if)# exit
```

```
Router5(config)# ip nat inside source static 192.168.25.2
    192.168.26.1
Router5# show ip nat translations
```

Listing 3.3: Static NAT on Router 5 (Area 3)

3.3.3 Translation Table (Verified)

Inside Local	Inside Global	Outside Local	Outside Global
192.168.19.11	192.168.20.5	192.168.20.5	192.168.20.5
192.168.19.12	192.168.20.5	192.168.20.6	192.168.20.6
192.168.19.13	192.168.20.5	192.168.20.7	192.168.20.7
192.168.26.1	192.168.20.2	—	—

3.4 ACL — Access Control Lists

3.4.1 What are ACLs?

ACLs are ordered permit/deny rules on router interfaces filtering traffic by source/destination IP or protocol — a stateless Layer-3 firewall.

Real-World Analogy: Airport Security

A security officer checks boarding passes — only valid ticket holders proceed through the gate. ACL checks each packet against rules and either allows or drops it.

Real usage: HDFC and **ICICI Bank** use extended ACLs so only authorised systems reach payment servers (PCI-DSS requirement).

3.4.2 ACL Implementation — Router 7

```
Router7(config)# ip access-list extended PROTECT_SERVER
Router7(config-ext-nacl)# 10 deny ip 192.168.1.2 0.0.0.0 any
Router7(config-ext-nacl)# 20 permit ip any 192.168.26.0 0.0.0.255
Router7(config-ext-nacl)# exit
Router7(config)# interface FastEthernet0/0
```

```
Router7(config-if)# ip access-group PROTECT_SERVER in
Router7# show access-list
```

Listing 3.4: Extended ACL protecting Area 3 server

3.4.3 Verification

Rule	Matches	Action
10 deny ip host 192.168.1.2 any	6	DENIED
20 permit ip any	15	PERMITTED

3.5 VPN — Virtual Private Network

3.5.1 What is a VPN Tunnel?

An IPSec VPN creates an **encrypted tunnel** over a public network so two sites communicate as if on the same private LAN. Implemented between **Router 1** and **Router 8**.

Real-World Analogy: Armoured Cash Van

Banks don't ship cash in open trucks — armoured vans with locked containers. VPN = your data travels the public internet inside an encrypted container only the two endpoints can open.

Real usage: TCS, Infosys, HCL — remote employees connect via IPSec/SSL VPN to corporate data centres. Mandatory Zero Trust requirement post-COVID.

3.5.2 Verification — Traceroute

```
C:\> tracert 192.168.100.2
  1   14ms  1ms  0ms  192.168.9.1
  2   15ms 10ms  2ms  192.168.100.2
Trace complete.
```

Listing 3.5: Traceroute confirming VPN tunnel

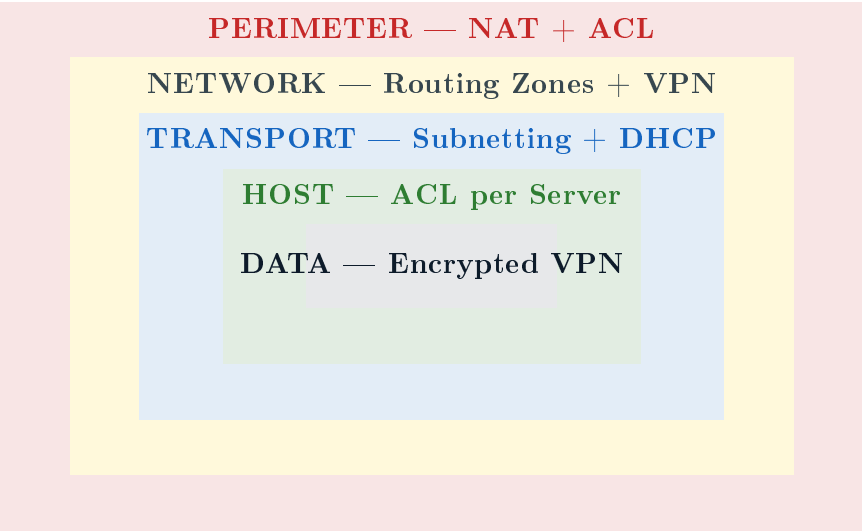
Key Insight

Only **2 hops** to the VPN destination confirms the IPSec tunnel encapsulates traffic correctly through the encrypted path, bypassing the open routing table.

4 Security Architecture Analysis

4.1 Defense-in-Depth Model

The GMNC network uses a **layered security model** aligned with **NIST CSF** and **ISO 27001**.



4.2 Threat Mitigation Summary

Threat	Mitigation	Mechanism
Unauthorized external access	NAT	Hides internal IP topology
Cross-zone server attacks	ACL	Denies inter-area server-to-server traffic
WAN data interception	IPSec VPN	Encrypted tunnel between Router 1 & 8
IP conflicts / rogue devices	DHCP	Controlled address pools per zone
Routing loops	RIP+OSPF	Metric-based convergence

5 Results & Conclusion

5.1 End-to-End Testing Summary

Feature	Test	Result
RIP Routing	`show ip route rip` — 30 networks	**PASS**
DHCP Auto-Assign	PC12 received 192.168.9.2 automatically	**PASS**
NAT Translation	Translation table populated	**PASS**
ACL Enforcement	Server blocked; PCs permitted	**PASS**
VPN Tunnel	Traceroute: 2 hops to 192.168.100.2	**PASS**
End-to-End Ping	All zones reachable	**PASS**

5.2 Conclusion

This project demonstrates a **scalable, secure, and fully functional enterprise network** for GMNC. Every component mirrors real-world deployments at Infosys, HDFC Bank, and Amazon AWS.

Key Achievements

- **30 subnets** routed via RIP v2 across 3 zones
- **DHCP automation** for 254+ devices per pool
- **NAT gateway** secures all internal areas behind one public interface
- **ACL policy** enforces zero-trust micro-segmentation at server level
- **IPSec VPN** provides encrypted site-to-site connectivity

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